

User Manual

Complete reference for all seventeen skills. Written against **gt v0.12.6**.

The model

Claude forgets everything between sessions. The vault doesn't. Golden Thread's job is to move what you learn *out* of a session into a file whose **location** tells you how widely it applies — then load only the slice the next session needs.

```
session conversation
  ↓ gt-work
Projects/<slug>/memory/*.md    this project, this detail
  ↓ gt-promote
research.md                    dated findings — `unverified` allowed, if labelled
  ↓ gt-promote + gt-validate
decisions.md · design.md       settled choices — independently verified
  ↓ gt-promote + gt-validate
Knowledge/                     true for several projects — independently verified
  ↓ gt-promote + gt-validate
global-memory/                 true everywhere — independently verified
```

Facts move **up** as they prove general. They never move back down, and nothing is silently deleted.

Verification gates the climb — it is not a step at the end. `research.md` holds dated findings and accepts an `unverified` figure so long as it is labelled. Every level above it *reads as settled*, so a figure entering `decisions.md`, `design.md`, `Knowledge/` or `global-memory/` — or driving a production change — must be **independently verified**: re-derived by a validator that never saw the reasoning. A second pass by the same session is only `self-verified`; it shares the assumptions that produced the error, so it confirms rather than checks. Run `/gt:gt-validate` *before* the promotion.

This rung was added after an unverified analysis reached `decisions.md` as an ADR recommending a four-ticker production change. An independent validator refuted three of the four — two of which would have deepened portfolio drawdown while appearing to improve their own ticker's. The ladder carried the error upward because nothing on it asked how the number was checked.

The discipline: write a fact at the narrowest scope that is honest, and promote it only when a second project proves it general. A fact promoted too early becomes a rule you must remember to disbelieve.

Loading is lazy, by design

`gt-open` reads the project's core docs and then reads `memory/MEMORY.md` **only** — an index of one line per file. The memory files themselves are read on demand.

A project with 70 notes costs ~80 lines to open instead of ~2,000. This is why `MEMORY.md` descriptions matter: they are the entire basis for deciding whether a file is worth opening.

Vault layout

```
<vault>/
  CLAUDE.md          vault-wide conventions and page schema
  index.md           navigational index of Knowledge pages
  log.md             audit trail
  review-queue.md    items flagged for owner review
  lint-declines.md   suppressions, each with its reason

  Sources/           IMMUTABLE raw originals
  Knowledge/         cross-project wiki pages
  global-memory/     loaded in every session
    MEMORY.md        the index – this is what actually gets read

  Projects/
    README.md        master list + Dataview views
    CONVENTIONS.md   stages, properties, domain taxonomy, file roles
    PROTOCOL.md      cross-project process rules
    INFRASTRUCTURE.md the server fleet, defined ONCE

<slug>/
  README.md          status board + property frontmatter
  source.md          where the code lives + deploy plan
  idea.md            original brain dump – IMMUTABLE
  research.md        append-only dated findings
  decisions.md       append-only numbered ADRs
  design.md          current architecture
  spec.md            handoff artifact (when design is complete)
  runbook.md         operational procedures (optional)
  memory/
    MEMORY.md        index – the only thing gt-open reads
  <topic>.md         loaded on demand
```

Project properties

Every project `README.md` carries frontmatter. These are real Obsidian properties — they drive the tag pane, search, and the Dataview views.

```
---
type: project
slug: my-project      # must match the folder name
domain: trading       # coarse grouping
stage: active         # idea|research|design|active|complete|archived
topology: bastion-direct # local|remote|bastion-jump|bastion-direct|n/a
tags: [trading, platform, live]
parent: other-slug    # sub-projects only
---
```

Categorise with properties, not folders. Do not add a category level under `Projects/`. `gt-lint` walks projects to find `memory/` directories; a category folder is fine for *that* check since v0.3.0, but folders also fragment `gt-open`'s project lookup and buy nothing Obsidian doesn't already give you through properties. The one legitimate second level is a real sub-project created with `--parent`.

Source topology

`source.md` records **where the code actually lives**. It is read first by `gt-open`, because acting without it is how you edit the wrong box.

TOPOLOGY	MEANING
<code>local</code>	A folder on this machine. No deploy step.
<code>remote</code>	One repo, one server. Repo mirrors the prod structure; the server pulls from GitHub.
<code>bastion-jump</code>	Several servers, all reached through one gateway host.
<code>bastion-direct</code>	Several servers, each reachable independently.

Addressing is **role × env → host + path**, because one machine can serve several vhosts and the same role exists in more than one environment.

Bastion topologies also carry a **file plan**, marking each deployed file:

- **static** — byte-identical on every target. Deploy from one canonical copy.
- **unique** — a per-server variant. Never copy one host's version over another's.

That distinction is the whole point. Two files with the same name on two hosts may be the same file that must not drift, or different files that must not be merged — and nothing but the file plan tells you which.

The fleet itself is defined **once** in `Projects/INFRASTRUCTURE.md` and linked from each `source.md`. Never copy the host table into a project.

Typical use cases

Recipes for the situations that actually recur. Each names the command; the per-command detail follows in the sections below.

Starting something new

```
/gt:gt-create → work → /gt:gt-work
```

`gt-create` gathers slug, domain, topology and tags, scaffolds the folder, and writes `idea.md` from what you actually said.

`idea.md` is **immutable after that** — it is the traceable "why", and the one file you never rewrite when the plan changes. Close the session with `gt-work` or the session's findings die with it.

Picking up a project after time away

```
/gt:gt-open <slug>
```

Reads the project docs in dependency order and stops at `memory/MEMORY.md`. Skim the returned summary for **stage**, **hosts** and **blockers** before touching anything — a stale host entry is cheapest to correct before work starts. Then pull individual memory notes by

name as the work needs them.

"What should I work on?"

```
python3 <vault>/Projects/golden-thread/tools/gt_tasks.py # then read TASKS.md
```

Regenerate first, always. Effective priority is computed against the clock — a `p:: 1` older than seven days escalates, a `due` inside three days escalates, an open `pp_escalate` window escalates. Reading a stale `TASKS.md` gives you last week's ranking with this week's confidence.

`waiting:: user` is your list; `waiting:: agent` is the session's.

Two sections sit above the ranked lists. **Inbox** shows every unchecked line in `INBOX.md` — thoughts captured from anywhere and not yet filed — so nothing is lost and nothing unfiled ranks. **Review** shows projects the close-out probe thinks may be finished, with the reasons, so closing a project is asked about rather than forgotten (see *Closing a project*).

Tasks at `p:: 7` or higher are **shelved**: kept in the README as a record, excluded from every section and every escalation rule, counted in *Project standing* only.

Bringing an existing project into the vault

```
/gt:gt-ingest
```

Copies — never moves or deletes. If the notes cross-reference each other by `[[wikilink]]`, **count the links before renaming anything**; Obsidian resolves links by filename, so a tidy-up destroys the graph. Prefer keeping filenames.

You just learned something — where does it go?

```
/gt:gt-work      (end of session)
/gt:gt-promote   (when it proves general)
```

File at the **narrowest scope that is honest**. A finding that stopped changing graduates from `memory/` to `research.md`; a truth a *second* project hit graduates to `Knowledge/`. See "Where does this fact go?" below. Promotion is cheap; demotion is not.

A number is about to be recorded as fact

```
/gt:gt-validate
```

Before it lands in `decisions.md`, `design.md`, `Knowledge/` or `global-memory/`, and before any production change. The validator gets the claim, the rules and the artifact — never your reasoning. Budget for the possibility that it comes back **refuted**; that is the skill working, not failing.

Answering "how does this work?"

```
/gt:gt-query <topic>
```

Reads `index.md`, follows wikilinks into `Knowledge/`, then falls back to grep and project memory. If a page comes back `status: stale`, verify before acting on it.

Keeping the vault honest

```
/gt:gt-upgrade      # after install.sh: migrate the vault (status, then --dry-run)
/gt:gt-doctor       # the whole install in one report; --fix re-wires hooks only
/gt:gt-lint         # broken links, orphans, unlisted memory, scope leaks
/gt:gt-runbook-lint # facts duplicated across runbooks
/gt:gt-refresh      # upstream changes to Sources/
```

Run `gt-lint` after any structural change. Triage into three piles: *you broke it* (fix now), *already broken* (record in `review-queue.md`), *false positive* (suppress **with the reason**). The check that matters most is `memory-unlisted` — a memory file missing from `MEMORY.md` is a file Claude will never load.

Capturing a thought without leaving what you are doing

Add one checkbox line to `<vault>/INBOX.md`, from any session, in any project:

```
- [ ] Backtesting: try the nq_pullback set against a cold cache
```

No project, no priority, no date. Then go back to work. Later:

```
/gt:gt-review
```

reads the inbox (and daily notes, if you keep them), asks where each line belongs, files it as a task or a project, and checks the inbox line off with a `→ [[slug]]` pointer. An idea filed as a task gets `p: 3` and no due date — a due date on an idea makes the deadline rule rank it above real work.

Closing a project

Delivery is not closure. A shipped project whose tasks stay open keeps escalating through its past due dates and outranks live work; on 2026-09-05 a talk delivered two days earlier held 24 of the top rows in the rollup.

```
python3 <vault>/Projects/golden-thread/tools/gt_closeout.py candidates
```

names projects whose signals say they may be finished: most open tasks past due, most tasks checked with nothing urgent left, three quiet weeks, or nothing open. The same probe runs at `/compact` and session end (setting `closeout_check`), and `gt-work` asks after the last urgent task is checked. The question is always put to you, never acted on.

Closing means: `stage: complete`, leftover tasks to `p:: 7` (shelved, not deleted), `pp: 3`, a last `research.md` entry, and a look at what to promote. Every ask and every answer is recorded in `Projects/golden-thread/closeout-signals.jsonl`; `gt_closeout.py history` shows what "ready" has actually looked like for you, which is what the thresholds get tuned against.

Setup

`/gt:gt-init`

First run, or wiring a new machine. Writes `~/.claude/vault-config.json` (the pointer every other skill reads), scaffolds the vault, and adds a Golden Thread section to `~/.claude/CLAUDE.md`. Also installs Core-rule enforcement hooks into `~/.claude/settings.json` (`UserPromptSubmit` + `Stop` hooks) so that Core rules are actively asserted, not merely stored. Idempotent.

For a vault that predates the Core-rule tier, run the install step separately:

```
python3 $SCRIPTS/vault_init.py install-core-rules --vault ~/my-vault
```

`CLAUDE.md` is read at **session start**. A rule added mid-session does not apply until you restart — that looks like the rule being ignored.

`/gt:gt-create`

Scaffolds a project. Gathers slug, title, tags, **domain**, sub-project parent, runbook, and **topology**; runs the script; then fills `idea.md` from what you actually said. `idea.md` is immutable afterwards — it is the traceable "why".

Writing to the shared files

`log.md`, `decisions.md` and `TASKS.md` are **generated**. They are the files every session writes and none owns, and in one working tree that means last-writer-wins with no conflict marker to warn you. So none of them is written directly.

```
T=<vault>/Projects/golden-thread/tools

# A log entry. Writes only this session's spool file, then re-renders log.md.
python3 $T/gt_log.py add "2026-01-01 10:00 CST [work] my-project - what happened"

# An ADR. Reserve the number FIRST -- two sessions that both read "the highest is 5"
# will both write ADR-6. This takes the number in a single atomic step and prints it.
python3 $T/gt_adr.py allocate my-project --title "The choice"
#   -> 7          (and creates the file that holds ADR-7; write the body into it)
python3 $T/gt_adr.py merge my-project

# Who has spooled what, and any allocation left unfinished
python3 $T/gt_log.py status
python3 $T/gt_adr.py status my-project
```

Adopting this in an existing vault is one command per file. It freezes what is there as a baseline that sorts first, so no history is rewritten and no ADR is renumbered, and it refuses unless the merge reproduces the original byte for byte:

```
python3 $T/gt_log.py migrate
python3 $T/gt_adr.py migrate my-project
```

`migrate` refuses on a project whose ADR numbers already collide, because renumbering would invalidate every existing reference to them. `gt-lint`'s `adr-collision` reports those so you can decide. A deliberate `## ADR-6 amendment:` is not a collision and is left alone.

Hand-editing a generated file is not a style violation — the change is lost at the next merge. `gt-lint` reports it as `generated-hand-edited`.

Both commands sit on one shared primitive, `gt_spool.py`: per-session spool files plus a merge whose ordering uses nothing machine-specific, so two machines rendering the same spools produce byte-identical output. You do not call it directly; it is documented here because a merge that ordered by mtime or directory order would make the *generated* file conflict in git, which is worse than the problem being solved.

Daily work

`/gt:gt-open <slug>`

Loads a project and stops. It is **read-only** — nothing is written during loading.

Reads, in order: `CONVENTIONS.md` and `PROTOCOL.md` (once per session), the project `README.md` — frontmatter first, since `domain / stage / topology / tags` is the fastest read of where the project stands — then `source.md` → `idea.md` → `research.md` → `decisions.md` → `design.md` → `spec.md` → `runbook.md`, the linked fleet page, and finally `memory/MEMORY.md` — the **index only**. Individual memory files load on demand.

`source.md` leads the project docs deliberately. It says which box serves which role in which environment. Acting before reading it is how you edit the wrong host, or overwrite a file that exists on three machines.

`research.md` over ~200 lines is read by its headings, plus the entries relevant to the work at hand and the most recent few. It is append-only and grows without bound; reading it whole crowds out the context the actual work needs.

Sub-projects load only when you name one, or the status board shows it as the active item.

Announces the `review-queue.md` count once, then summarises stage, **topology and hosts**, next action, blockers, and what memory exists but is unloaded — the host list so you can correct a stale entry *before* work starts rather than after.

/gt:gt-route

The middle of a session. `gt-open` runs before the session knows what it is; `gt-work` runs after it is over. Between them sits the part where the work actually changes shape, and nothing served it.

Run it whenever the ground has shifted, or whenever you cannot name what you are doing. It reads only the conversation already in context — never project files, because reading `research.md` to decide where a note goes costs more than the note — and answers four things:

1. **What this has become.** Restated in a sentence, with the drift named explicitly if the session no longer resembles what it opened with. This is most of the value: the drift is usually felt before it is articulated.
2. **Where the output belongs** — the [Where does this fact go?](#) table, applied to the material actually in hand, naming more than one destination when more than one is true.
3. **Whether the work is in the right place** — project loaded, harness (a real PTY belongs in a terminal; vault context belongs in Claudian), and whether the model matches the difficulty. It reports mismatches; it changes nothing.
4. **One next action.** A skill to invoke or a file to append to — not a menu.

It writes **nothing**, with a single exception: one `INBOX.md` checkbox line, which is what `INBOX.md` is for. Everything else is handed to the skill that owns the destination, which does the writing under its own claim.

"Nothing, keep going" is a valid answer and the skill is built to give it. A check that always finds work is a check nobody runs twice.

Use it when a session opened as one thing and became another; when a finding might belong to a project other than the one loaded; when you are about to write something down and are not sure which file; or when you simply feel lost. Being wrong costs one re-run.

/gt:gt-work

End of session. The step people skip, and skipping it is what makes the vault decay.

GOES TO	WHAT BELONGS THERE
<code>research.md</code>	Dated findings and gotchas. New discoveries only.
<code>decisions.md</code>	Numbered ADR: decision, context, rejected alternatives. Stable choices only.
<code>design.md</code>	Rewritten in place if architecture changed. Describes <i>now</i> .
<code>memory/*.md</code>	Session state, updated in place.
<code>spec.md</code>	Created when design is complete enough to hand off.

Memory files carry `level` and `enforcement` in frontmatter to make a rule's durability explicit:

```
metadata:
  type: core | feedback | user | reference
  level: core | context | generic      # default: generic
  enforcement: validated | reminder    # required only for level: core
```

Default to `generic` . Do not set `level: core` here — Core is a promotion, and it requires wiring an enforcement hook, which is `gt-promote` 's job.

After writing, `gt-work` actively asks whether any finding applies beyond this project and should go to `Knowledge/` , `global-memory/` , or a new project.

Also updates the `stage:` , `topology:` , and `domain:` properties in `README.md` when the project's reality changes.

Not in `decisions.md` : anything you might change next session. An ADR you reverse next week teaches the vault to lie to you.

`/gt:gt-query`

Reads `index.md` , follows wikilinks 2–3 hops into `Knowledge/` , falls back to grep, then project memory. Be specific. If a result is `status: stale` , verify it.

`/gt:gt-ingest`

Bulk-imports an existing project. Copies — never moves or deletes.

The trap: if the notes cross-reference each other by `[[wikilink]]` , renaming or merging files destroys that graph, because Obsidian resolves links by filename. Measure first:

```
grep -roh '\[([^\]]*)\]' --include='*.md' . | wc -l
```

Prefer keeping filenames. If you must repoint, match the *whole* link — `[[name]]` and `[[name|alias]]` — never a substring, or `[[foo]]` will corrupt `[[foo_bar]]` . Never repoint links inside `Sources/` .

`/gt:gt-review`

Empties `INBOX.md` : reads every unchecked line (and daily notes, if configured), asks where each belongs, files it as a task or project, and checks the inbox line off with a pointer. Regenerates the rollup at the end.

Context management

`/gt:gt-farm`

Routes work out of this context to an external AI service when work is bulk, mechanical, or genuinely benefits from a non-Claude second opinion.

The unit is a **work packet**: self-contained, with a strict return contract (`FINDINGS` + `GAPS` blocks). The transport is swappable — the user pastes it into a web UI today; a script sends it to an API later. The packet is identical either way.

All four gates must pass before a task leaves. Any failure means the work stays here.

GATE	QUESTION
Stateless	Answerable with no vault or repo state?
Self-contained	Fits in a paragraph plus a list of URLs?
Checkable	Verifiable without redoing the work?
Releasable	Every input safe to hand a third party?

Releasable is default-deny. Never include: account identifiers, balances, hostnames, internal IPs, credential locations, non-public source code, or personal identifiers.

Key rules: - One packet, one task. A packet asking three questions returns three half-answers. - Supply a vocabulary block with exact model IDs, product names, and version numbers read from the system — never from memory. - Results come back as **unverified**. Promote a finding only after opening its **source:** URL and confirming the claim. - **GAPS** is mandatory and **GAPS: None** must be justified, not asserted. - Cite the specific page carrying the claim, never a homepage.

Packets are saved to `<vault>/Projects/external-ai-tools/packets/<YYYY-MM-DD>-<slug>.md`.

Knowledge management

/gt:gt-promote

The judgement call: **has a second project proved this general?**

FROM → TO	WHEN
memory/ → research / decisions	The finding stopped changing session to session
project → Knowledge/	A second project hit the same truth
Knowledge/ → global-memory/	Needed in every session, every project
anywhere → new project	The idea doesn't belong where it is

Nothing is deleted. Retiring sets **status: stale** and records what superseded it.

Promoting a rule to Core (**level: core**) also wires the enforcement hook — **gt-promote** handles this; **gt-work** does not.

/gt:gt-refresh

Checks **Sources/** for upstream changes. Supersedes with a **new** immutable file carrying **supersedes:** — never edits the old one.

Verification

/gt:gt-validate

Verify a claim by **re-deriving** it, not by reviewing it. Use before a finding is recorded as fact, before a production change, or whenever a number matters.

The one rule: the validator never receives the reasoning that produced the claim. Give it the reasoning and it grades the argument — inheriting the same blind spot. Give it only the claim, the rules and the artifact, and it has to go back to primary sources. That is the only

thing that catches a wrong premise.

Reduce the work to a **single falsifiable assertion**. "The analysis is sound" is not validatable; "widening NQ's target to 1.25 improves risk-adjusted return" is. Validate several claims **separately** — a bundled claim returns a bundled verdict, which hides which part failed.

Loads the project's `validation-rules.md`, and the parent's too, since packs are inherited. Those are standing invariants the validator enforces whether or not the request mentions them — because *a requester who has already made a domain error will not think to ask the validator to check for it*. On conflict, the pack wins.

CLASS	USE WHEN THE RISK IS
empirical	A number, measurement or result could be wrong
vantage	The measuring position may not be able to observe the answer
rule-compliance	Work must satisfy <code>core-rules/</code> or <code>CONVENTIONS.md</code>
code	Code may not do what its name, comment or docs claim

Pick every class whose risk is present. **Do not substitute one generic reviewer** — that produces agreement, not verification.

The packet is exactly three fields — `claim`, `rules`, `artifact` — then re-read it and strip your conclusions, your numbers, your confidence, and any adjective implying the expected answer. "Confirm that X" becomes "determine whether X". Withholding *reasoning* is isolation; withholding *access* is just a broken validator, so keep the hostnames, API shapes and filters it needs to reach the artifact alone.

Run the validators **independently**, and read their derivation *before* re-reading your own — reading yours first turns the exercise back into review.

Three verdicts: **confirmed**, **refuted** (quantify the divergence), and **cannot-verify** (name what was missing).

Append the verdict to `research.md`. A refutation supersedes the original finding.

cannot-verify is never a pass. A validator that could not check something and stayed quiet manufactures false assurance — worse than no validator at all.

You are the worst possible author of this packet, because you already know the answer and your framing leaks. Treat building it as an adversarial exercise against yourself. A validator disagreeing is a *result*, not a failure.

Maintenance

/gt:gt-lint

CHECK	CATCHES
index-gap	Knowledge page not in index.md – gt-query won't find it
broken-link	[[wikilink]] resolving to nothing
orphan	Knowledge page linked from nowhere
memory-unlisted	memory/*.md not in MEMORY.md – Claude will never load it
global-gap	global-memory/*.md not in its index
memory-bloat	global-memory file over 30 lines
global-scope-leak	Project-specific content in global-memory/
superseded-cited	Knowledge page citing a superseded source
stale	Knowledge page marked status: stale
source-todo	source.md with no topology or deployment targets
frontmatter	Project README missing properties, or slug ≠ folder
core-misplaced	A rule declares level: core but lives outside core-rules/
core-no-enforcement	level: core with no enforcement field declared
core-unenforced	The declared enforcement hook is not actually wired

core-unenforced is the critical one — it is the machine-checkable form of "rule stored but never asserted". Treat it as a real defect. Do not suppress it — suppressing core-unenforced recreates the original bug with a paper trail saying it was fine. Fix: vault_init.py install-core-rules --vault <vault> .

Declines go in lint-declines.md as suppress: with the reason.

Triage findings into three piles: you broke it (fix now), already broken (record in review-queue.md — don't guess at a target), false positive (suppress with reason).

/gt:gt-settings

View and change what Golden Thread does automatically. Every automatic behaviour is registered here and can be switched off.

SETTING	VALUES	DEFAULT	WHAT IT DOES
<code>component_updates</code>	<code>off</code> · <code>report</code> · <code>confirm</code> · <code>auto</code>	<code>report</code>	At session start, compares installed hooks/scripts against plugin source and reports drift
<code>version_check</code>	<code>off</code> · <code>report</code>	<code>report</code>	At session start, reports when a newer plugin version is checked in than the one installed
<code>orphan_check</code>	<code>off</code> · <code>report</code> · <code>reap</code>	<code>report</code>	At session start, looks for abandoned background Claude workers; <code>reap</code> stops them
<code>push_check</code>	<code>off</code> · <code>report</code>	<code>report</code>	At session start, reports vault commits not yet pushed
<code>report_card</code>	<code>off</code> · <code>minimal</code> · <code>full</code>	<code>minimal</code>	At <code>/compact</code> , summarises session hygiene
<code>watch</code>	<code>off</code> · <code>report</code>	<code>off</code>	<code>/gt:gt-watch</code> : the hourly fetch and the session-start report of repo changes
<code>test_gate</code>	<code>off</code> · <code>warn</code> · <code>auto</code> · <code>block</code>	<code>auto</code>	Refuse a <code>git commit</code> of code whose tests have not been seen to pass; <code>auto</code> blocks only where the repo has a test command
<code>parallel_work</code>	<code>off</code> · <code>on</code>	<code>on</code>	Whether divisible work runs in parallel at all; <code>off</code> also stops the Core rule being injected
<code>parallel_max</code>	<code>auto</code> · a positive integer	<code>auto</code>	Ceiling on concurrent workers. <code>auto</code> = as many as the machine allows
<code>install_demo</code>	<code>yes</code> · <code>no</code>	<code>yes</code>	Whether <code>install.sh</code> installs <code>/gt:gt-demo</code> , its script and the PizzaBot template

Heads-up: `parallel_work` is `on` and `parallel_max` is `auto` out of the box, so after upgrading to 0.12.4 a long run will use as much of your machine as it can — many processes at once, CPU well above 100%, and the fans to match. That is intended. Cap it with `set parallel_max <N>` or switch it off with `set parallel_work off`; nothing else about the plugin changes.

Test receipts. `core_test_before_commit` needs evidence that the tests passed on *this* tree, so a run leaves a receipt and the commit guard reads it:

```
gt_test_receipt.py record --repo . --what "pytest -q" --ok # any project
gt_test_receipt.py latest --repo . # what was last recorded
gt_test_receipt.py check --repo . --files a.py b.py # does a receipt cover these?
```

A receipt covers a file only if it is newer than that file, so editing something after the run invalidates it automatically. `tests/run.sh` and `dev/release-check.sh` record their own. A repo with no tests is exempted once with `touch .gt-no-test-gate`; a single commit with `GT_TEST_GATE=off`.

The machine profile. `parallel_max: auto` resolves through `parallel_profile`, which `install.sh` measures on this machine and re-measures at every upgrade — cores for CPU-bound work, 2× cores (bounded by memory) for I/O-bound. Inspect or refresh it:

```
gt_settings.py detect-machine # what this machine would measure
gt_settings.py detect-machine --write # measure and store it
```

Setting a number above what the machine can use is refused at the point you set it, and a number above core count is accepted with a note that it only helps I/O-bound work.

The budget as a number. Any script, in any project, can ask what the settings allow instead of inventing a worker count:

```
JOBS=$(python3 ../scripts/gt_settings.py jobs 42 --io-bound) # 42 units of I/O-bound work
JOBS=$(python3 ../scripts/gt_settings.py jobs 8)             # 8 units, CPU-bound
```

It returns 1 when `parallel_work=off`, never more than `parallel_max`, and never more workers than there are units of work. `gt_settings.parallel_jobs()` is the same answer from Python.

```
python3 $SCRIPTS/gt_settings.py show          # current state of all settings
python3 $SCRIPTS/gt_settings.py explain <name> # reasoning behind the default
python3 $SCRIPTS/gt_settings.py set <name> <value>
```

`component_updates` at `auto` will not overwrite an installed file that is newer than the source, and will not delete a file the source lacks — real drift ran that direction, so a naive "source is truth" updater would silently break enforcement on every session start.

`/gt:gt-runbook-lint`

Finds procedures duplicated across runbooks and routes them to the right shared layer — `PROTOCOL.md`, a Knowledge page, or a repo `CLAUDE.md`.

`/gt:gt-watch`

Watch any git repo you depend on, and hear about it when it changes in a way you said you care about — raised as a **P0** when it ships a security fix.

```
/gt:gt-watch add <git-url>    - create a watch (any git URL: GitHub, GitLab, self-hosted, file://)
/gt:gt-watch list             - every watch and its last change
/gt:gt-watch check            - fetch now instead of waiting for cron
/gt:gt-watch show <slug>      - Claude reads the new commits and explains them
/gt:gt-watch ack [<slug>]     - mark changes seen
/gt:gt-watch remove <slug>    - stop watching
```

A watch is a note — `Projects/golden-thread/watches/<slug>.md`, editable in Obsidian's Properties panel. Its frontmatter says what to watch: `url`, `track` (commits, tags, releases), `branch`, `watch_paths`, `current_version`, your own `p0_when` patterns, and `starred`.

How it runs. A cron job (`gt_watch.py install-cron --every 1h`) fetches every watch without Claude, classifies each change, and queues it. At session start a hook reads the queue — P0s first, spelled out; review items one line each; routine changes as a count. It never fetches. Machine state lives in `~/ .claude/golden-thread/watch/`, outside the vault.

When it is a P0 — decided by rules, never by a model reading commit prose: a new security advisory for the repo; a new tag or release naming a CVE or GHSA id, or saying "security fix/release/update" or "vulnerability"; a commit message carrying a CVE or GHSA id; a change to `SECURITY.md` or your `watch_paths` that says the same; or anything matching your own `p0_when`. A new release, a

major-version jump, a change under `watch_paths`, or "BREAKING CHANGE" is a **review** item. Ordinary words like "auth" or "security" in a commit message are not enough — a P0 that cries wolf stops being read.

Off until you switch it on: setting `watch ( off · report )`. `add` offers to switch it on and install the cron entry.

`/gt:gt-demo`

An eleven-act guided tour of Golden Thread on PizzaBot 3000, a fictional pizza-ordering project. It runs in its **own throwaway vault**, so nothing it does can reach yours.

```
/gt:gt-demo start    - build the demo vault and print the command that opens it
/gt:gt-demo tour     - (in the demo session) run the tour; you only click Next
/gt:gt-demo end      - every commit and file the tour produced
/gt:gt-demo clean    - delete the demo vault and build a fresh one
/gt:gt-demo remove   - delete the demo and switch it off (install_demo = no)
/gt:gt-demo status   - is there a demo vault, and how old is it
```

Running it: `start` builds the vault at `~/.claude/golden-thread/demo-vault` and prints `cd <demo vault> && GT_VAULT=<demo vault> GT_WATCH=report GT_WATCH_STATE=<demo vault>/.demo/watch claude` — the extra variables let the watch act run without touching this machine's watch settings or state. Run that in a new terminal and type `/gt:gt-demo tour`. For each act Claude says one line to the audience, does the work with the real skill, says what just happened, and offers buttons — **Next**, **Repeat this act**, **Skip ahead**, **End tour**.

The acts: Core rules enforced (a prepared reply carrying a key is blocked) · open a project · "what's next?" from the task rollup · capture a finding and an ADR mid-session · ingest a source and query the wiki · route a stray idea · validate a wrong claim · lint finds a planted broken link · promote a finding to a Knowledge page (which fixes the link) · watch an upstream library and see its security release open as a P0 · file the inbox, close the session, and show the receipt. The acts are plain text in `templates/demo-pizzabot/tour.md` — reorder or reword them there.

Why its own vault: a fuller tour writes to shared files (`INBOX.md`, `TASKS.md`, `log.md`, `Knowledge/`) that other sessions also write. Undoing that in a real vault means rewinding history everyone shares. In a throwaway vault, `clean` just rebuilds it. The demo never writes your `vault-config.json`: the demo session is pinned with `GT_VAULT`, which every skill and hook honors. `clean` and `remove` delete a directory only if it carries the demo's marker file.

Where does this fact go?

THE FACT	GOES TO
"This preset is wrong right now"	memory/
"This API rejects plural keys"	research.md
"We use POST for all mutations, and why"	decisions.md
"The system is structured like this now"	design.md
"This platform behaviour bit two projects"	Knowledge/
"Always diff before overwriting prod"	global-memory/
"How to restart the prod backtest"	runbook.md
"Never scp without backing up"	PROTOCOL.md
"This code lives on host X at path Y"	source.md
"Which box serves which role"	INFRASTRUCTURE.md

Two failure modes, opposite directions: **too narrow** buries a platform truth in one project, so the next rediscovers it the hard way; **too wide** promotes a quirk into `global-memory/`, where it loads into unrelated sessions as a rule that isn't true there. When unsure, file narrow — promotion is cheap, demotion isn't.

/gt:gt-route applies this table for you, against the material actually in hand, without your having to hold the table in your head mid-session. Reach for it when the answer is not obvious — including when the honest answer is that the fact belongs to a project other than the one loaded.

Script reference

```
SCRIPTS=~/.claude/plugins/cache/golden-thread-plugin/gt/0.12.6/scripts

python3 $SCRIPTS/vault_init.py fresh --vault ~/my-vault --domain "My Team"

python3 $SCRIPTS/vault_init.py create-project --vault ~/my-vault \
  --name my-project --title "My Project" --tags "backend,api" \
  --domain platform --topology bastion-direct \
  --repo-url git@github.com:me/my-project.git --runbook

python3 $SCRIPTS/vault_init.py create-project --vault ~/my-vault \
  --name sub-feature --parent my-project --domain platform

python3 $SCRIPTS/vault_init.py connect --vault ~/existing-vault

python3 $SCRIPTS/vault_init.py install-core-rules --vault ~/my-vault

python3 $SCRIPTS/gt_ingest.py ~/Projects/my-project --json

python3 $SCRIPTS/gt_lint.py ~/my-vault --queue ~/my-vault/review-queue.md

python3 $SCRIPTS/gt_settings.py show
```

`gt_tasks.py` runs from the vault, at `Projects/golden-thread/tools/`, beside `gt_closeout.py`, `gt_session.py`, `gt_edits.py` and `safe_write.py`. All five ship as templates and are seeded there by `vault_init.py` `fresh` or `connect` (what `/gt:gt-init` runs); `install.sh` refreshes copies that predate the installed template. It regenerates `TASKS.md` from every project's `## Tasks` section.

```
python3 ~/my-vault/Projects/golden-thread/tools/gt_tasks.py
```

It infers the vault from its own location, so it needs no arguments; `--vault` is there to override that. **Re-run it before reading `TASKS.md`** — project priority is computed against the clock (stale-P1 ageing, deadline windows, `pp_escalate`), so the ranking changes with time even when no file has changed. `TASKS.md` is generated and must never be hand-edited.

Troubleshooting

"vault-config.json not found" — run `/gt:gt-init`.

A rule in `CLAUDE.md` seems ignored — it is read at session start. Added mid-session, it applies from the *next* session. Same for hooks in `settings.json`: open `/hooks` once or restart.

Memory files aren't being read — that is intended. `gt-open` reads `MEMORY.md` and loads files on demand. Ask for one by name.

A memory file is never offered — run `/gt:gt-lint`; it is probably `memory-unlisted`, so nothing knows it exists.

`/gt:gt-query` can't find a page you added — likely `index-gap`.

Lint suppression isn't working — paths are matched case-insensitively against the finding path; make sure you used the path as the finding reports it.

"conflict: vault-config.json points to a different vault" — run `vault_init.py connect --vault <new-path>`.

core-unenforced finding in lint — Core rules are stored but not being asserted. Run `vault_init.py install-core-rules --vault <vault>` or wire the hooks by hand. Do not suppress this finding.